

Supplementary Information

Understanding epistatic networks in the B1 β -lactamases through coevolutionary statistical modeling and deep mutational scanning

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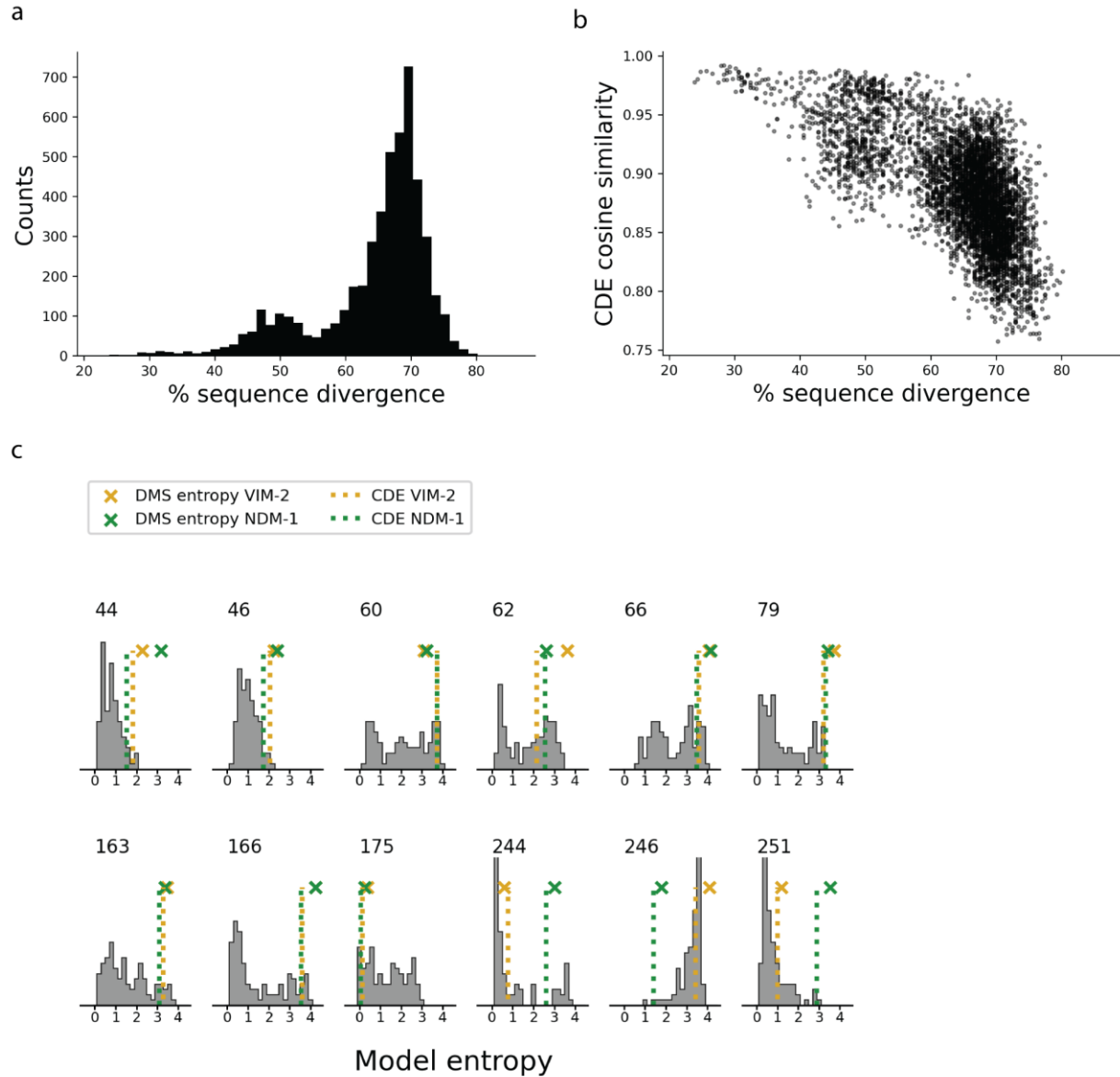
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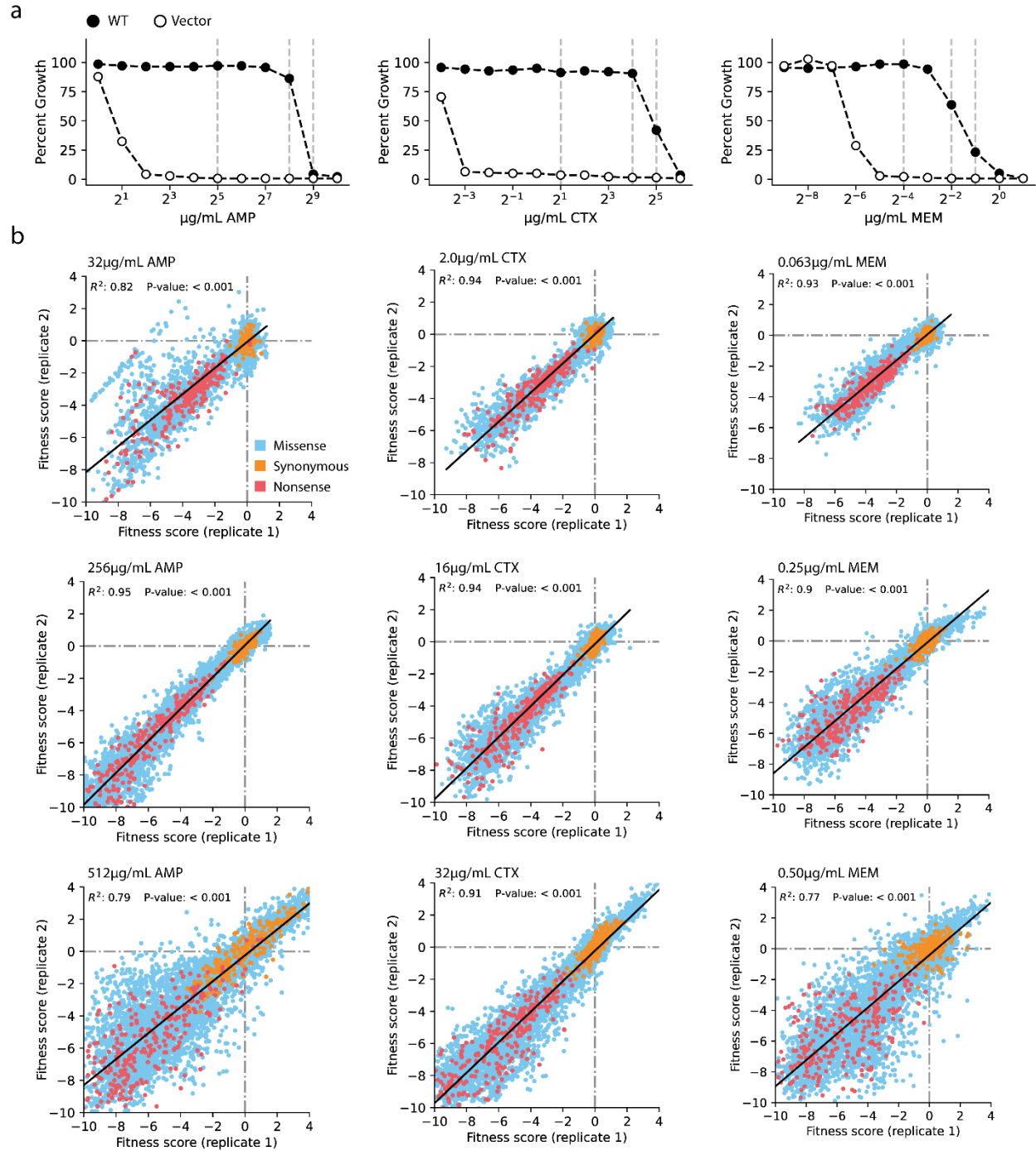
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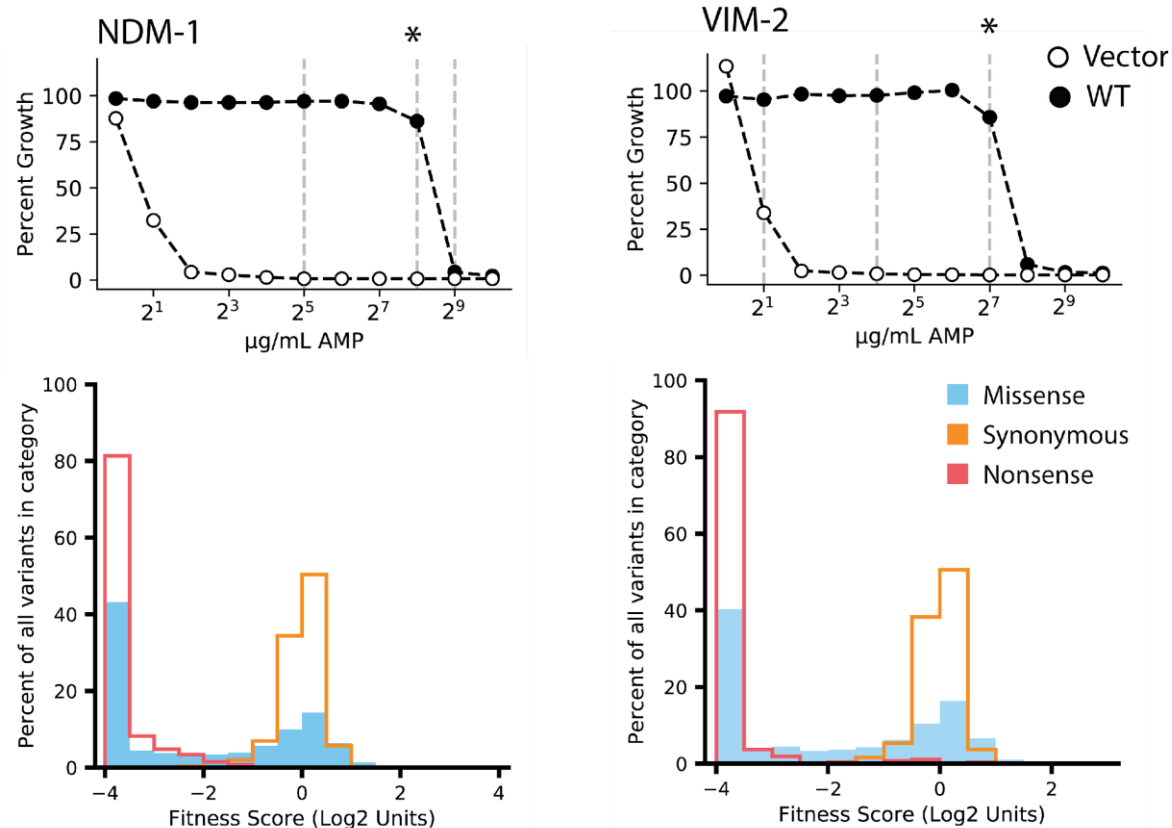
Supplementary Figures



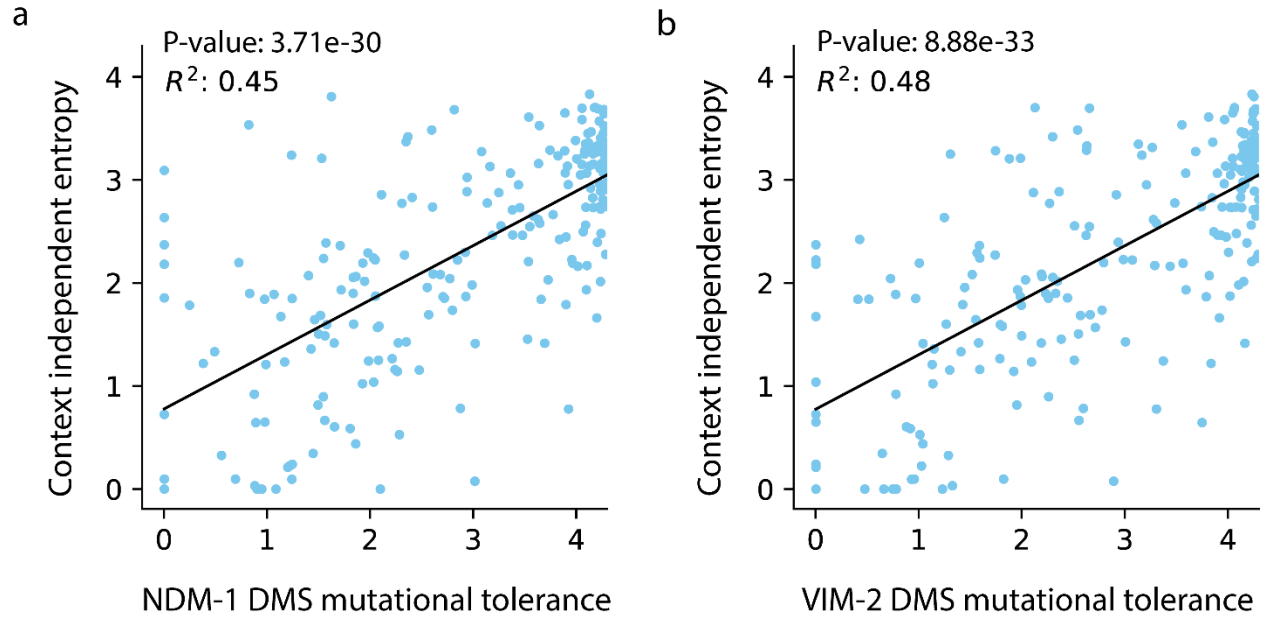
Supplementary Figure 1. DCA reveals sequence and position specific variation in mutational tolerance across B1 MBL homologs. (a) Pairwise percent sequence divergence between all pairs of the 100 selected homologs. (b) Plot of percent sequence divergence versus cosine similarity between CDE (predicted mutational tolerance) for all pairwise comparisons of 100 homologs. The cosine similarity captures the similarity of mutational tolerance across all aligned positions (encoded as a vector) of each pair of homologs being compared. (c) Distributions of CDE across the 100 homologs for 12 different positions throughout the proteins. The selected positions have been specifically chosen to showcase instances where DMS and DCA predictions are in good agreement for NDM-1 and VIM-2. The specific values for NDM-1 and VIM-2 (later tested, but not part of the 100 homologs) with both DCA and DMS entropies are highlighted in green and yellow. Positions are labeled by the VIM-2 positions.



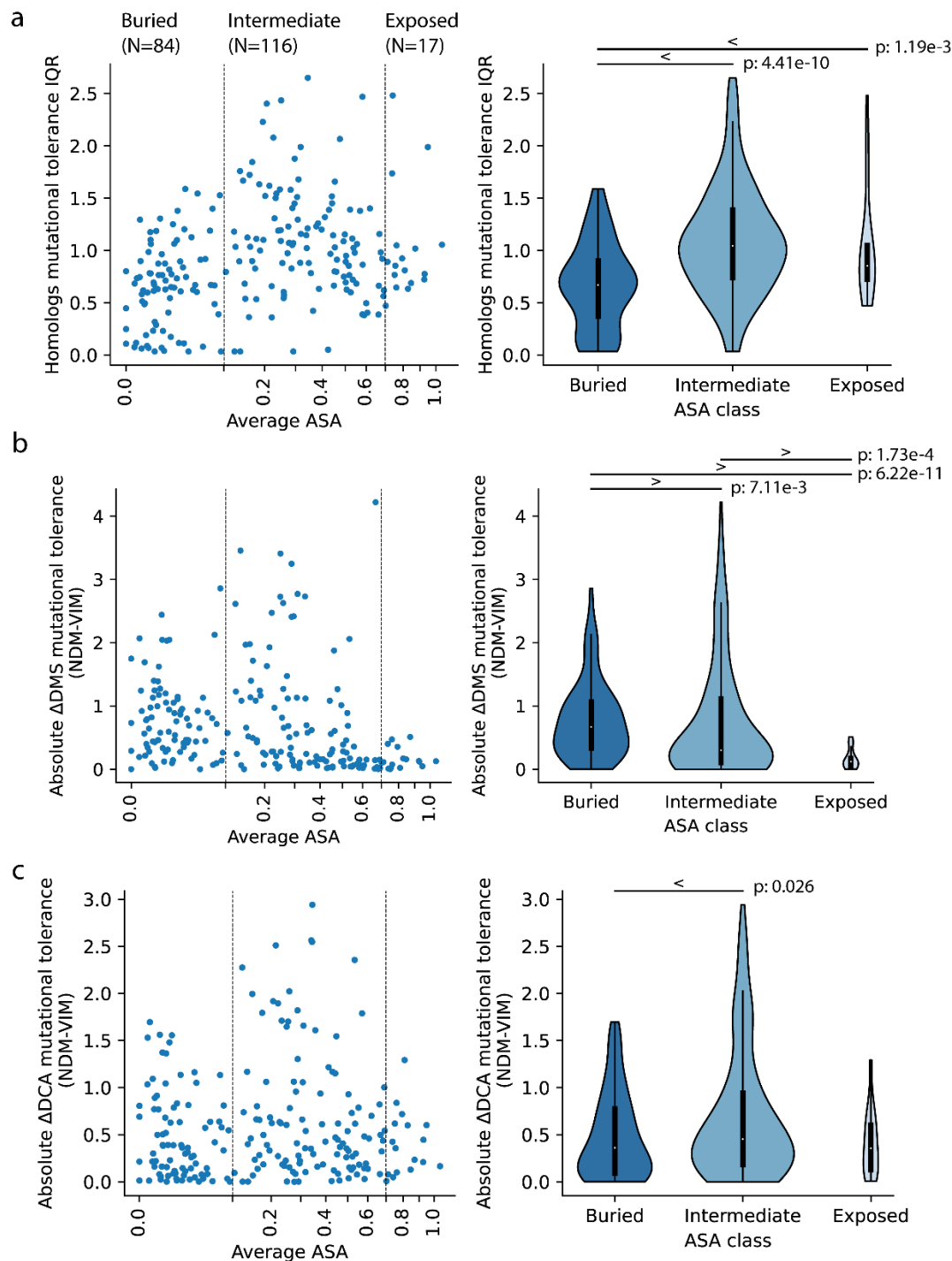
Supplementary Figure 2. Selection range and replicate correlation of NDM-1 DMS experiments. (a) Growth curves for wtNDM-1 and empty vector under selection with 3 different antibiotics. Vertical dashed lines indicate conditions explored in DMS experiments. (b) Replicate correlation for NDM-1 DMS data, conducted in the 3 antibiotics at 3 concentrations each.



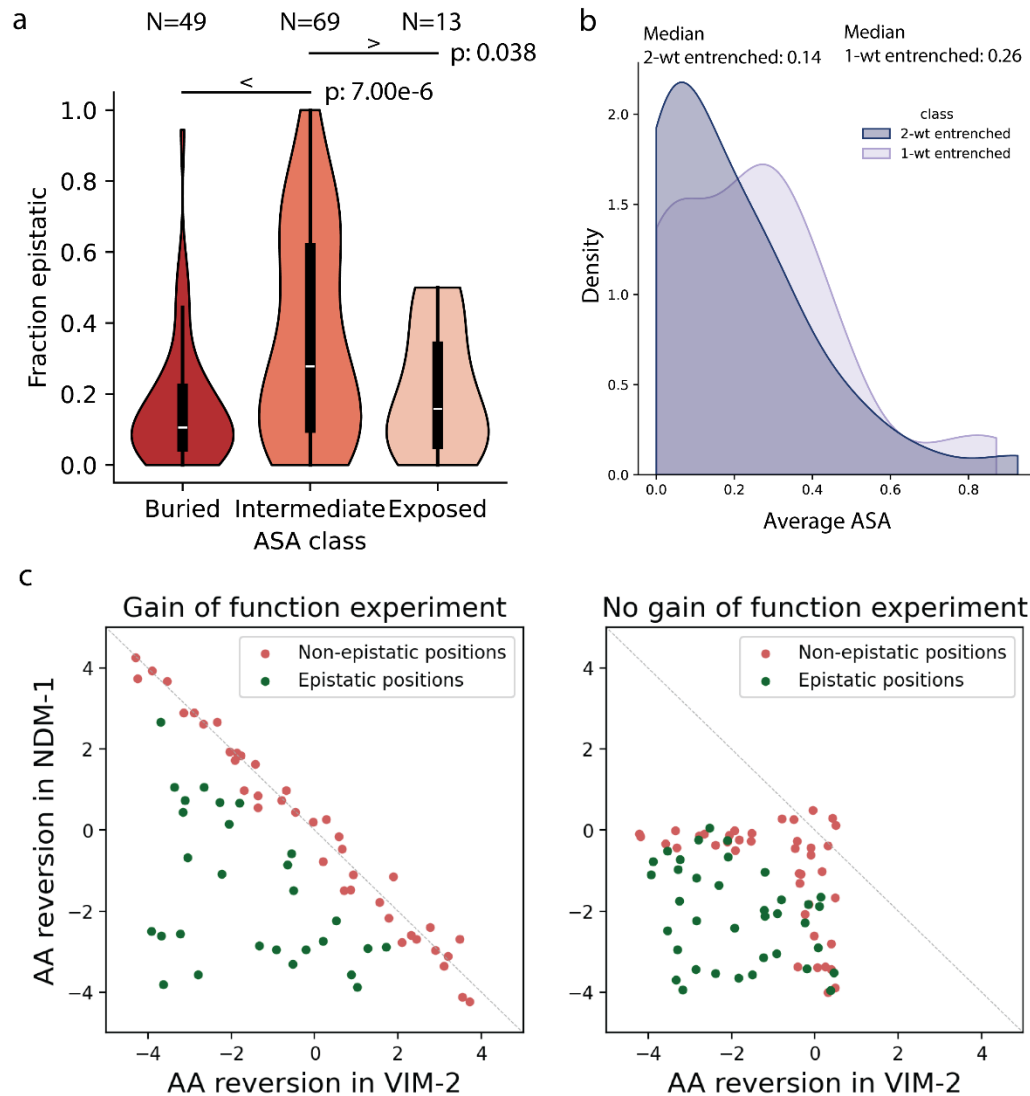
Supplementary Figure 3. Comparison of selection pressures between NDM-1 and VIM-2 dataset. In the top panels, the dose-response curve shows the survival of *E. coli* expressing each WT or an empty vector across increasing selection pressures of AMP. The stars indicate the selection conditions of the datasets being compared directly in this study. The bottom panels show the distribution of fitness effects of all mutants of NDM-1 or VIM-2, split into categories by the type of mutation.



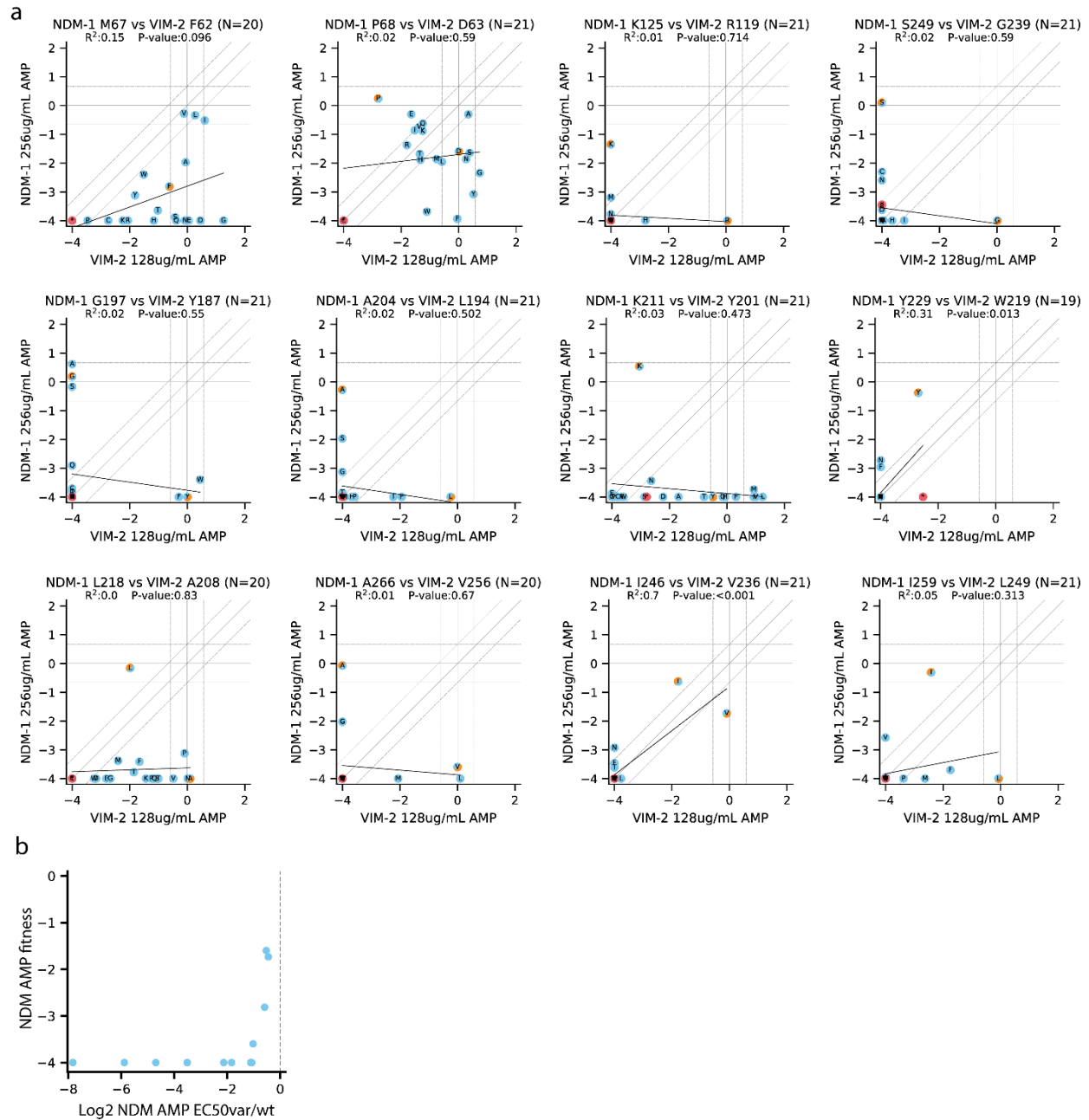
Supplementary Figure 4. Correlation between mutational tolerance observed in DMS and context independent entropy. Context independent entropy (CIE) is calculated as the Shannon entropy of the per position amino acid frequencies for each aligned position in the model, without considering coupling to other positions in the sequence background. **(a)** CIE vs NDM-1 DMS mutational tolerance. **(b)** CIE vs VIM-2 DMS mutational tolerance. The line of best fit and R^2 of a linear regression are shown.



Supplementary Figure 5. Variation in mutational tolerance compared to residue burial. The measures of variation in mutational tolerance at each position being examined are (a) the IQR of DCA mutational tolerance across 100 homologs, (b) the difference in DMS mutational tolerance of NDM-1 and VIM-2, and (c) the difference in DCA mutational tolerance between NDM-1 and VIM-2. Scatter plots of variation in mutational tolerance vs average ASA are on the left, with regions of different levels of burial (ASA class) highlighted (buried: ASA < 0.1, intermediate: $0.1 \leq \text{ASA} < 0.7$, exposed: ASA ≥ 0.7). The distribution of each ASA class with regards to variation in mutational tolerance is plotted in the violin plots on the right (area scaled by data count). Lines at the top indicate where values in one distribution tend to be larger than the other based on a one-tailed Brunner-Munzel test (p -value < 0.05), with the direction of the inequality indicated above the lines.



Supplementary Figure 6. Fraction of epistatic mutations and epistasis of entrenched residues. (a) Distribution of the fraction of epistatic mutations at shared positions, split by level of residue burial (buried: $ASA < 0.1$, intermediate: $0.1 \leq ASA < 0.7$, exposed: $ASA \geq 0.7$). Violin plots are scaled by data count. Lines at the top indicate where values in one distribution tend to be larger than the other based on a one-tailed Brunner-Munzel test (p -value < 0.05), with the direction of the inequality indicated above the lines. (b) Distribution of average ASA for positions that are 1-WT or 2-WT entrenched. (c) Hypothetical distribution of reversion mutations in an experiment capable of detecting gain of functions (left) or incapable of detecting gain of functions (right).



Supplementary Figure 7. Fitness scores at positions in the NDM-1 background selected for mutation to the VIM-2 WT. (a) Scatter plots show the fitness scores of mutants between NDM-1 and VIM-2 at aligned positions (labeled at top of each figure). Points with orange color in one half indicate the synonymous mutant for the background (e.g., orange in lower right side indicates a synonymous variant of VIM-2); some synonymous variants are not plotted (M, W have only one codon), but the mutation of the other background to the WT are still in the dataset. Red points are nonsense variants. Blue points indicate all other missense variants. Vertical and horizontal dashed lines indicate the regions of neutral effects based on the synonymous variant distributions of each homolog. The diagonal dashed lines have the same region of neutral effects as the y-axis. **(b)** EC50 vs DMS fitness of the selected single mutants. EC50 of each variant is expressed as the fold change from wtNDM-1.